Huawei Access Node BAS-ACC-07 Hardware/Environmental Fault Resolution SOP

Node: BAS-ACC-07 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_6827

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve hardware/environmental faults on the Huawei Access Node BAS-ACC-07 at the Basavanagudi POP, specifically addressing ASIC memory parity errors, line card crashes, and power supply redundancy loss. The Access Node is a critical component in the Huawei network architecture, responsible for providing high-speed, high-capacity access to the network core. The Node's Line Card (LC) interfaces with the Physical Layer (PHY) and Data Link Layer (DLL) protocols to establish and maintain network connectivity. In the event of a hardware or environmental fault, the Node may generate an Alarm Code (ALARM_6827) indicating a Minor failure. This SOP aims to identify the root cause of the failure, perform diagnostic tests, and implement corrective actions to restore normal operation.

Pre-requisite Checks

- Verify the Basavanagudi POP's environmental conditions, including temperature, humidity, and power supply status.

- Check the redundant link status and ensure it is operational.

- Dispatch remote hands to the Basavanagudi POP for physical access and assistance.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to manufacturing defects or wear and tear.	High
Line card crashes caused by software bugs, configuration issues, or hardware malfunctions.	Medium
Power supply redundancy loss due to hardware failure or improper configuration.	Low

Diagnostic Commands & Expected Outputs

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Command: display fault-management alarm, Expected Output: ALARM_6827: Minor failure, Description: This command displays the current alarm status, including the Alarm Code (ALARM_6827) and severity (Minor).

Command: display hardware, Expected Output: ASIC memory parity errors detected, Description: This command displays the Node's hardware status, including any detected errors or malfunctions.

Command: display power status, Expected Output: Redundant power supply lost, Description: This command displays the Node's power status, including any loss of redundancy.
```

Resolution Steps

Step 1: Step: Replace faulty ASIC, Command: huawei> replace asic 1/1, Description: This command initiates the replacement of the faulty ASIC with a new one.

Step 2: Step: Reboot the Node, Command: huawei> reboot, Description: This command restarts the Node to ensure the new ASIC is recognized and functioning correctly.

Step 3: Step: Verify power supply redundancy, Command: display power status, Description: This command verifies that the redundant power supply is operational and functioning correctly.

Post-Check Verification

```
Command: display fault-management alarm, Expected Output: No alarms, Description: This command verifies that the alarm status has returned to normal.

Command: display hardware, Expected Output: No errors detected, Description: This command verifies that the Node's hardware status has returned to normal.
```

Roll-back Procedures

- Step: Restore original ASIC, Command: huawei> restore asic 1/1, Description: This command restores the original ASIC in case the replacement failed.

- Step: Reboot the Node, Command: huawei> reboot, Description: This command restarts the Node to ensure the original ASIC is recognized and functioning correctly.